

Ex.no:7b	DATA VISUALIZATION USING TABLEAU (E-COMMERCE ANALYTICS)

Aim:

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective E-Commerce business intelligence and data analytics.

Algorithm:**Step 1: Data Loading**

- Connect the E-Commerce dataset to Tableau.
- Verify the imported fields such as product_name, category, country, unit_price, quantity, and order_date.
- Load the dataset and verify the data types of the imported fields.

Step 2: Product Quantity Bar Chart

- Create a worksheet for product-wise quantity.
- Place product_name on Columns.
- Place quantity on Rows.
- Select the Bar Chart visualization.
- Enable data labels to display the quantity of each product.

Step 3: Category-wise Quantity Visualization

- Create a worksheet using category and quantity.
- Aggregate quantity using SUM.
- Visualize the contribution of each category using a Pie Chart.
- Display category names and corresponding quantities as labels.

Step 4: Purchase Trend

- Create a Line Chart using order_date and quantity.
- Place order_date on Columns and quantity on Rows.
- Aggregate quantity using SUM.
- Use the chart to observe the purchasing trend over time.

Step 5: Purchase Value

- Create a calculated field named Purchase Value.
- Use the formula:

[unit_price] * [quantity]

- Display product_name, quantity, unit_price, and Purchase Value in a worksheet.

Step 6: Country Filter

- Add country to the Filters shelf.
- Select a country from the filter.
- Observe the updated product quantities and purchase values based on the selected country.

Step 7: Minimum Quantity Parameter

- Create a parameter named Minimum Quantity.
- Set the required minimum quantity value.
- Apply the parameter to display only products whose quantity meets or exceeds the selected value.

Step 8: Dashboard Creation

- Create a new Tableau Dashboard.
- Add the Product Quantity, Category Distribution, Purchase Trend, and Purchase Value visualizations.
- Arrange the worksheets clearly.
- Add the Country filter and Minimum Quantity parameter.
- Format the dashboard with suitable titles and labels.

Step 9: Story Creation

- Create a Tableau Story using the completed dashboard.
- Add story points for:
 1. Overall E-Commerce Performance
 2. Category-wise Analysis
 3. Product-wise Analysis
 4. Final Purchase Analysis
- Present the visual insights using the story.

Step 10: Analysis

- Analyze the charts to identify the highest and lowest-selling products.
- Compare quantities among different product categories.
- Observe purchase trends over time.
- Analyze purchase value across products and countries.
- Observe how the Country filter and Minimum Quantity parameter dynamically change the visualizations.
- Use the dashboard and story to summarize the overall E-Commerce performance.

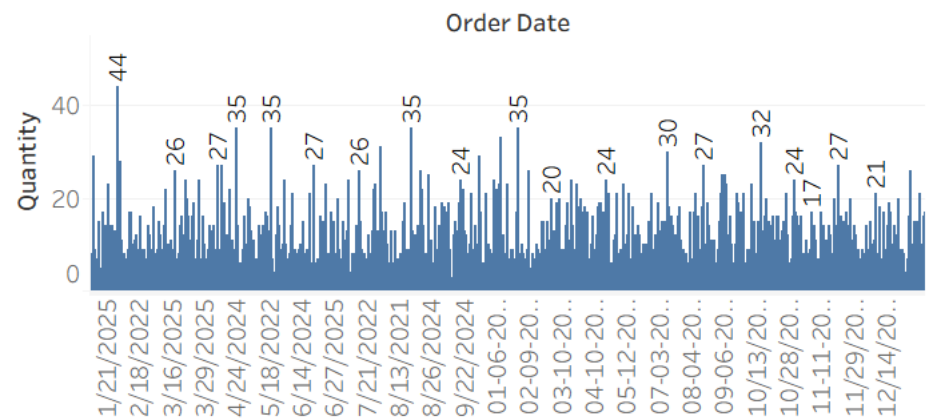
Output:



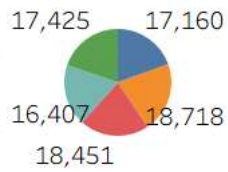
Category wise unit price



Purchase Trend



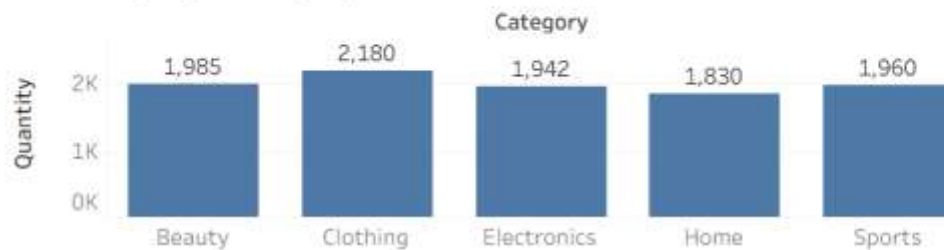
Category



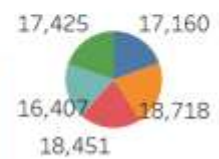
E-Commerce Analytics Dashboard

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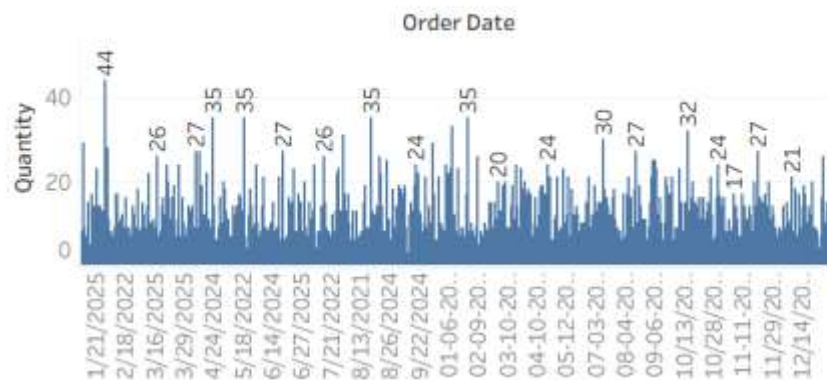
Quantity by Category



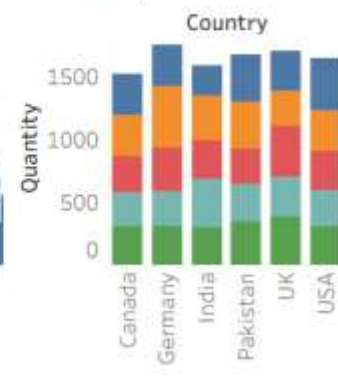
Category



Purchase Trend



Country wise



Result: